Homework Assignment for Week 7

[Due time and date: 11:00 pm on 29 March, 2026]

Topic: Trade with Price Action and AI Pattern Recognition, by Dr. Larry Cheung

Instruction:

1. **Cover Page Requirements:** Your submission must include a cover page. Clearly list the following details: (1) Group Number; (2) Full names and student ID numbers of your group members.
2. **Formatting & Submission:** All solutions must be typed. Submit a single document per group to the "**Homework Assignment for Week 7**" folder on Blackboard. **Importantly, t**o ensure compatibility with AI grading tools, *start the solution for each new question on a separate page*. Include page numbers throughout the document.
3. **File Format:** Please submit your file in .docx (Word) format. Alternatively, .pdf or .odf files are accepted. No other formats will be processed (by AI); specifically, please do not submit legacy .doc or .wps files!!!
4. **Collaboration Advice:** While this is a group assignment, I strongly recommend that every member attempts the problems independently. Then your team members meet for discussions and work out a better solution for submission. Doing so will significantly improve your preparation for the final exam.

Question 1: In the context of the trader's equation, how does the relationship between entry timing, risk-reward ratio (RR), and probability typically behave, and why is this critical for quantitative model design?

Question 2: The presentation argues against the efficacy of isolated, traditional candlestick patterns (e.g., Harami, Shooting Star). From a quantitative and price action perspective, what is the fundamental flaw in relying solely on these patterns, and what alternative approach is suggested?

Question 3: When applying Convolutional Neural Networks (CNNs) to recognize k-line (candlestick) features, what specific data augmentation techniques are appropriate, and which should be strictly avoided? Why?

Question 4: Why should technical analysis work?

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